CyberDefense Pro 2.0

2.5.8 Maintenance Windows

This lesson covers the following topics:

- Understand maintenance windows
- Analysis of events
- Patch management

Understand Maintenance Windows

Many organizations adopt routine maintenance windows so administrators can perform maintenance tasks during these pre-established times. Maintenance windows enable preventative maintenance and consistent deployment of noncritical patches. All work planned during maintenance windows should comply with change management policies. Computers and devices are often restarted during maintenance windows, and various services are also modified, restarted, and added. Monitoring infrastructure must be able to correlate events like these to a scheduled maintenance window to adjust alert severity ratings.

Maintenance tasks typically fall into one of two categories:

Category	Description
Proactive	Proactive maintenance is designed to prevent future issues or safely perform work that may impact system performance. Maintenance windows are generally associated with preventative maintenance tasks, as reactive maintenance typically cannot be delayed.
Reactive	Administrators perform reactive maintenance in response to a problem or an outage.

Analysis of Events

Analysis of events occurring during maintenance is essential, as a savvy attacker might try to avoid detection by performing actions during these times. Knowing what will happen in a maintenance window is critical to help discern between authorized and unauthorized events. A poorly planned and managed maintenance window can be a nightmare for the SOC as it tries to manage a sudden surge in alerts and warnings generated by numerous services as they are added, modified, and restarted.

Patch Management

Patch management teams rely on maintenance windows to complete patch rollouts. The duration of the maintenance window can be a significant constraint.

Change management policy dictates that patching must finish quickly enough to accommodate rollback plans if trouble occurs without overrunning the maintenance window. Change management rollback is the process of undoing a system's changes to restore the system to an earlier, pre-change state.

Rollbacks can be performed manually or automatically, depending on the system, and are done to return a system to its previous state. The need for a rollback is not always obvious. Even after passing initial tests, some changes introduce issues that may not become apparent until a system is subjected to heavy workloads or scenarios that are not easy to test.

After changes have been made to a system, analysts must monitor it to verify that it is operating as expected. Monitoring often involves log monitoring (looking for errors or warnings) and performance monitoring (comparing characteristics against an established baseline).